ME750: Sheet Metal engineering Report on
Design, Development, and Analysis of Tractor Foot Step Using Auto Form Software



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Abstract

This project focuses on the design, development, and analysis of a tractor foot step using Auto Form software to ensure ergonomics, durability, and manufacturability. The 3mm-thick sheet metal component was evaluated through simulations for formability, thinning, spring back, and wrinkles, with results confirming no defects (tears/splits), acceptable thinning (-0.2 to 0.2), minimal spring back (±0.5mm), and uniform strain distribution (major strain: 0.263; minor strain: 0.172). The design meets all functional requirements, offering anti-slip properties, weldability, and corrosion resistance, and is ready for prototype validation.

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Introduction

1.1 Overview

This report documents the design, simulation, and analysis of a tractor foot step using Auto Form software, focusing on sheet metal forming principles to achieve an ergonomic, durable, and manufacturable component. The study addresses critical engineering challenges in automotive part development, including material selection (3mm thickness), anti-slip requirements, and prevention of forming defects, while leveraging computational simulation to optimize the manufacturing process. By analyzing formability, thinning, spring back (±0.5mm), and strain distribution (major strain 0.263, minor strain 0.172), the project demonstrates how digital validation can replace traditional trial-and-error methods, reducing development time and costs. The findings not only validate the foot step design but also provide a replicable framework for similar medium-volume sheet metal components in agricultural equipment manufacturing, emphasizing first-time-right production and quality assurance in emerging market applications

1.2 Part detail and application

The 3mm thick sheet metal foot step provides safe, ergonomic access for tractor operators with its anti-slip design. Engineered for durability, it offers excellent weldability, formability (13mm height), and corrosion resistance. The component withstands agricultural demands through optimized strength-to-weight ratio and manufacturability via stamping processes. Its design ensures operator comfort while meeting rugged field requirements, balancing structural integrity with production efficiency for farm equipment applications.

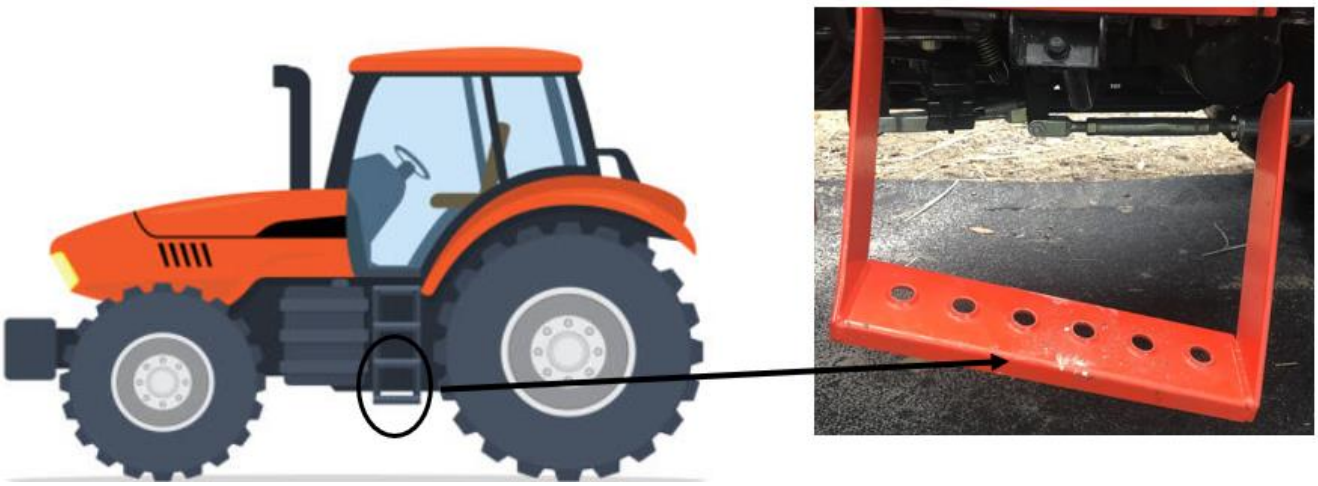


Figure 1 – Foot step on tractor

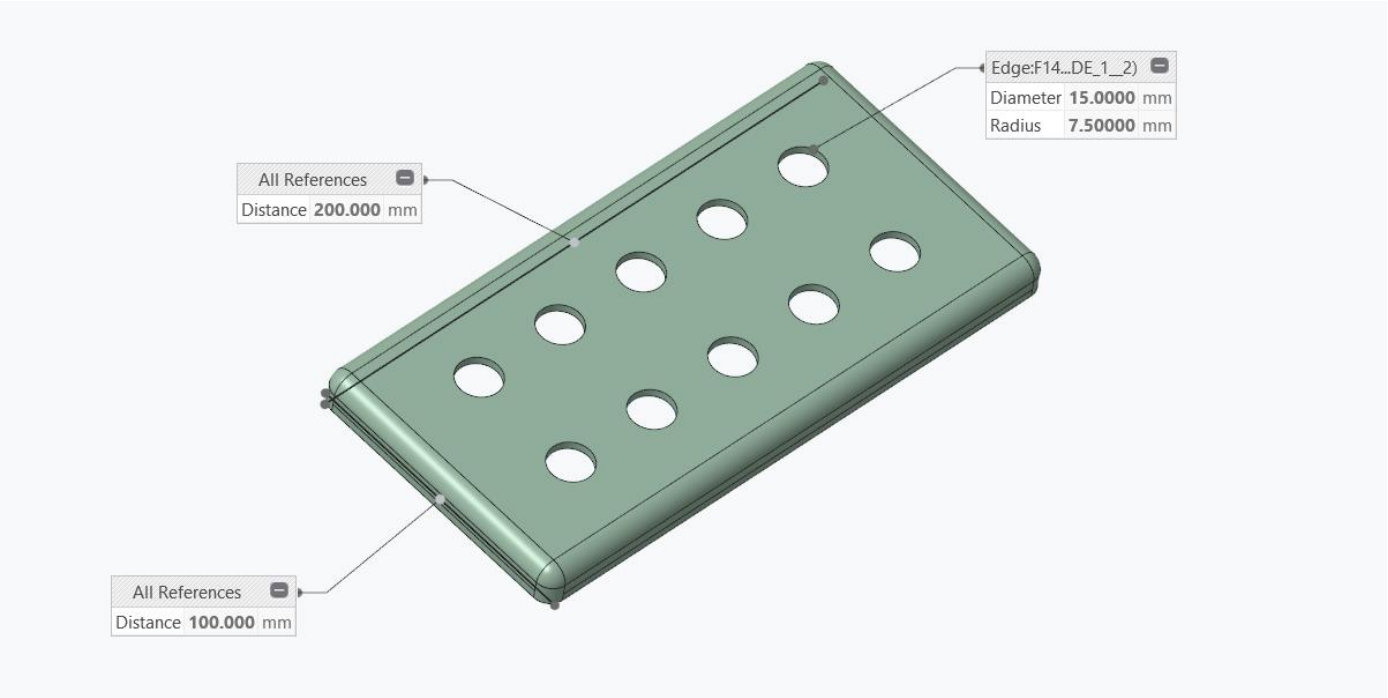


Figure 2 – 3D Part created in Creo Software

1.3 Material Properties

The foot step is fabricated from **Non-alloy Structural Steel (S355)** compliant with **ISO 1052** standards, offering an optimal balance of strength and manufacturability for sheet metal applications.

Parameter	**Value**
Material Standard	**ISO 1052 (Steel for General Engineering Purpose)**
Material Used	**Non-alloy Structural Steel (S355)**
Tensile Strength	**470 - 630 MPa**
Yield Strength	**≥ 355 MPa (340-440 MPa)**
Density	**7.85 g/cm³**
Poisson's Ratio	**0.3**
Modulus of Elasticity (E)	**210 GPa**

- High Strength & Durability:
- Good Weldability:
- Moderate Corrosion Resistance:
- Machinability & Formability: Easily cut, drilled, and machined
- Good for sheet metal operations.

2.Autoform Analysis

2.1 Auto form process flow

The Auto Form simulation process for the tractor foot step involved a systematic approach to validate formability and manufacturability. First, the 3D CAD model was imported into Auto Form, followed by defining the material properties (S355 steel) and process parameters. The simulation setup included tool design (punch, die, binder), blank positioning, and lubrication conditions. Key analyses such as formability (FLD), thinning, spring back, and wrinkle risk were performed to evaluate the design.

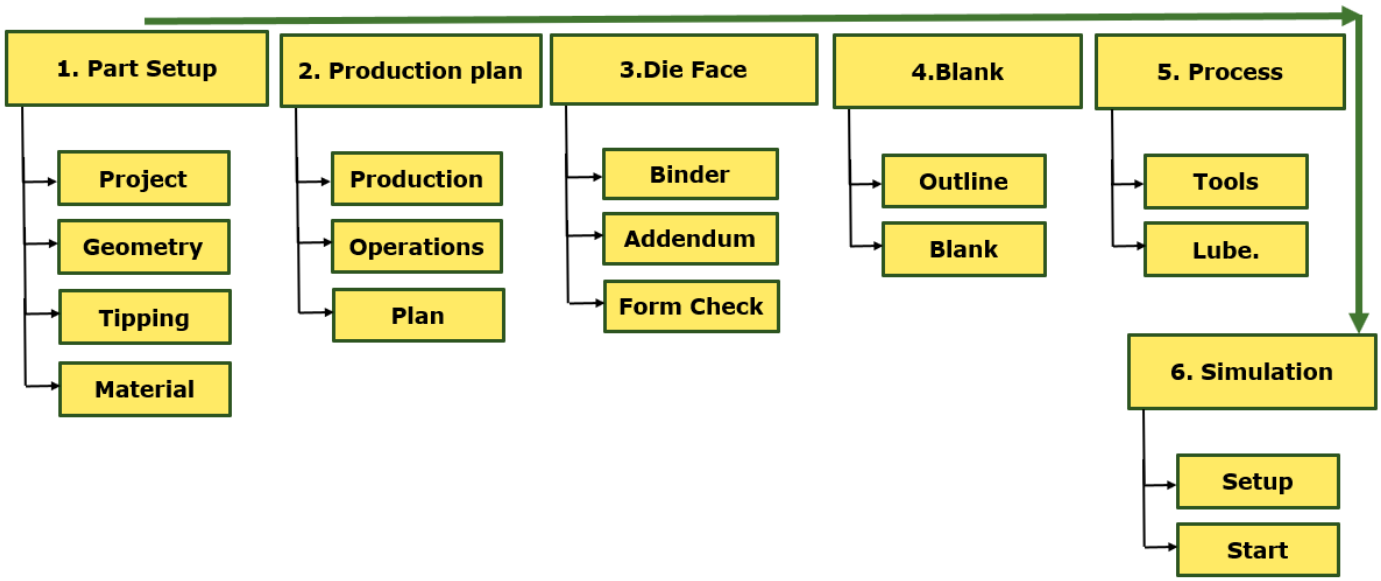


Figure 3 – Auto form Process Flow

2.2 Punch, Die and Binder preparation in Auto form

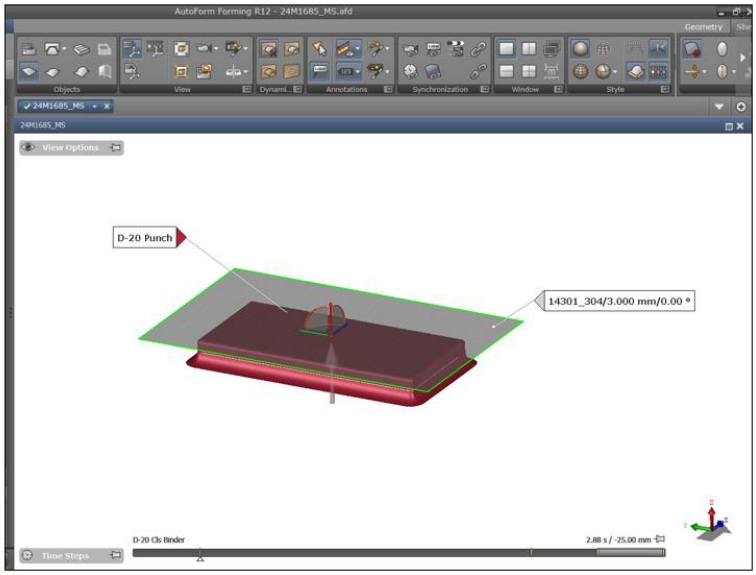


Figure 4 – Punch and blank formation

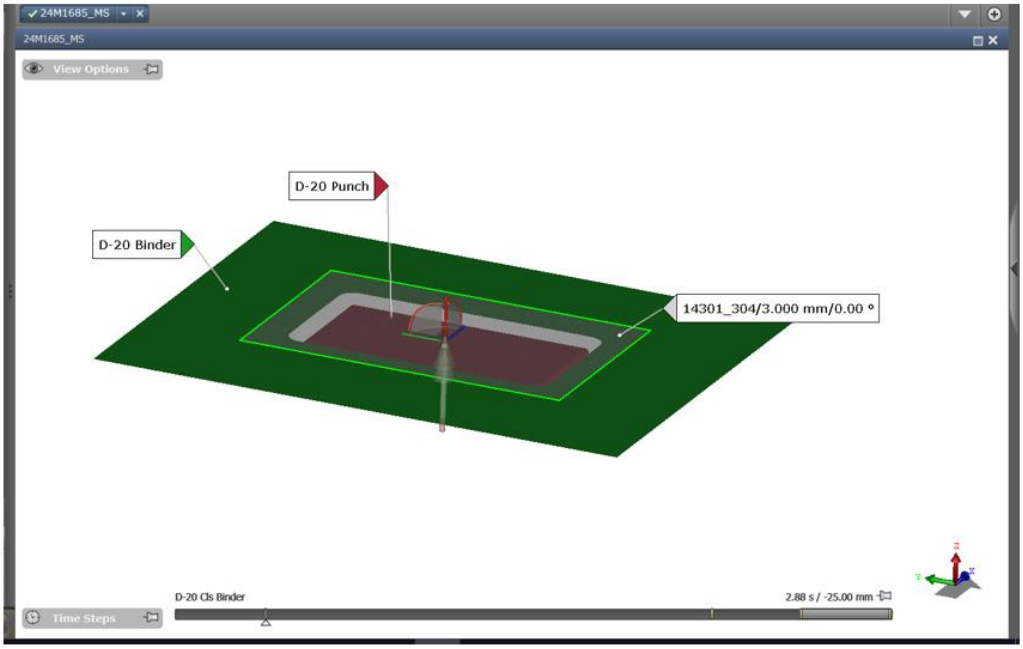


Figure 5 - Binder

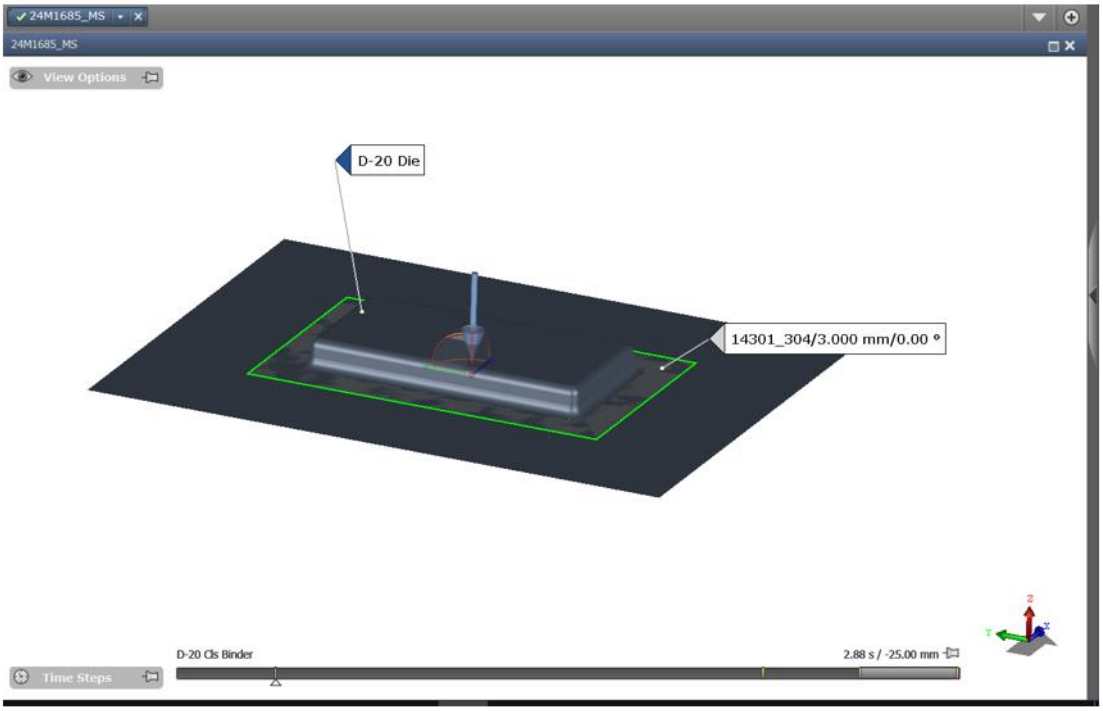


Figure 6 - Die

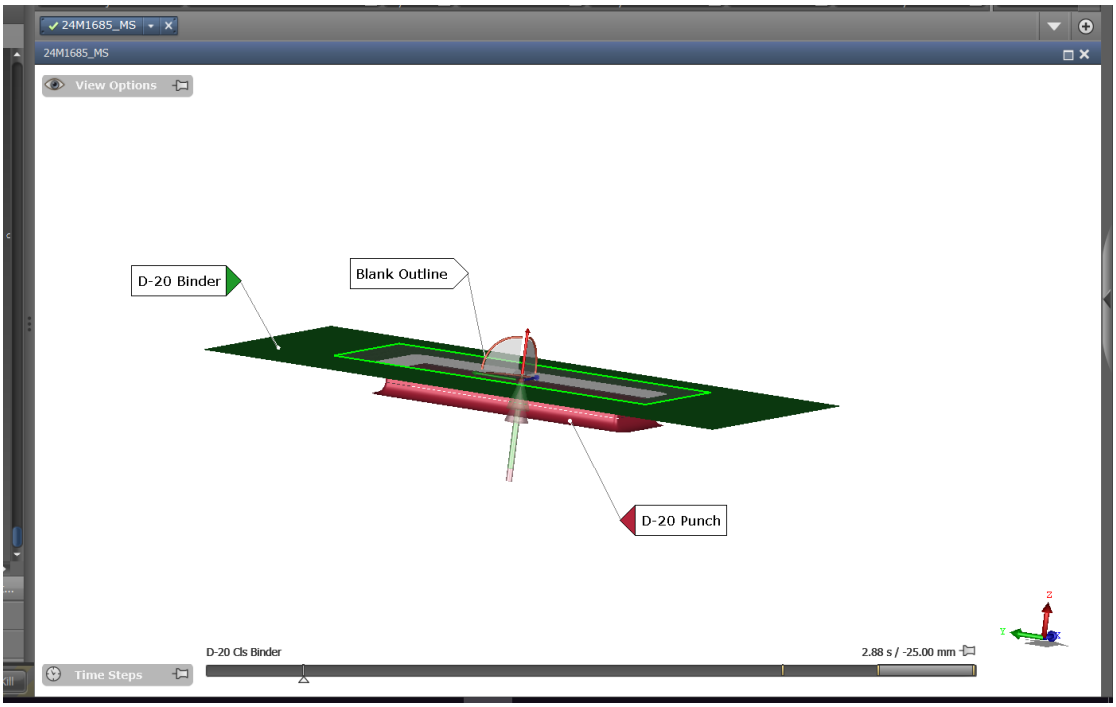


Figure 7 – Lower Forming Tool

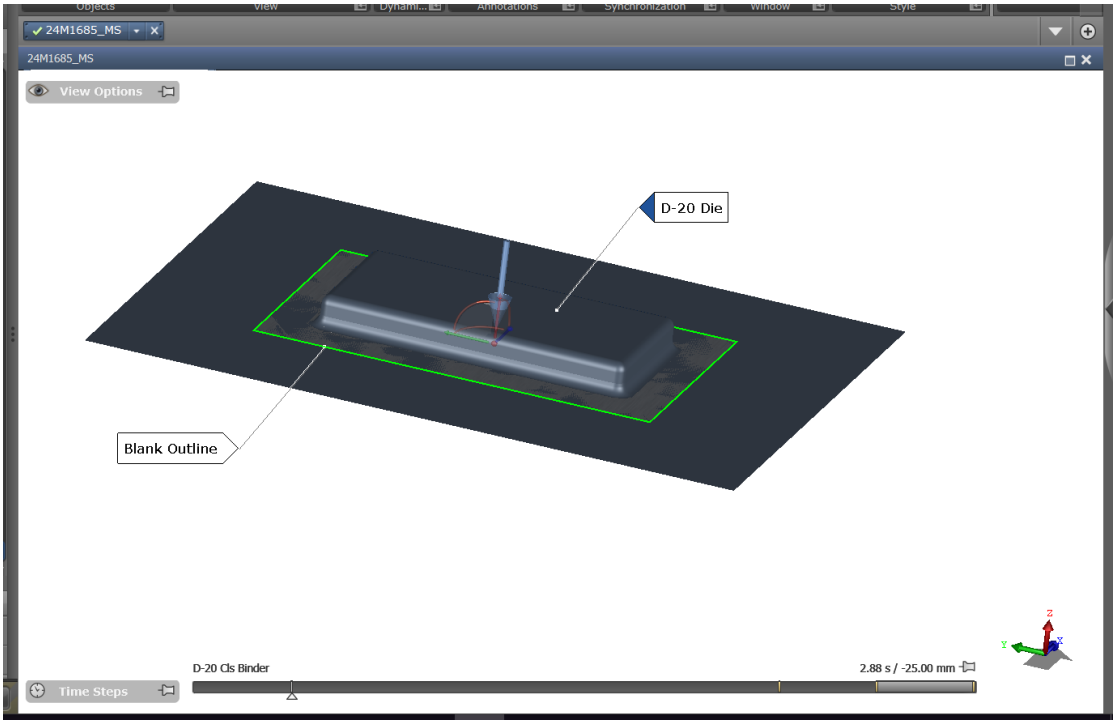


Figure 8 – Upper Forming Tool

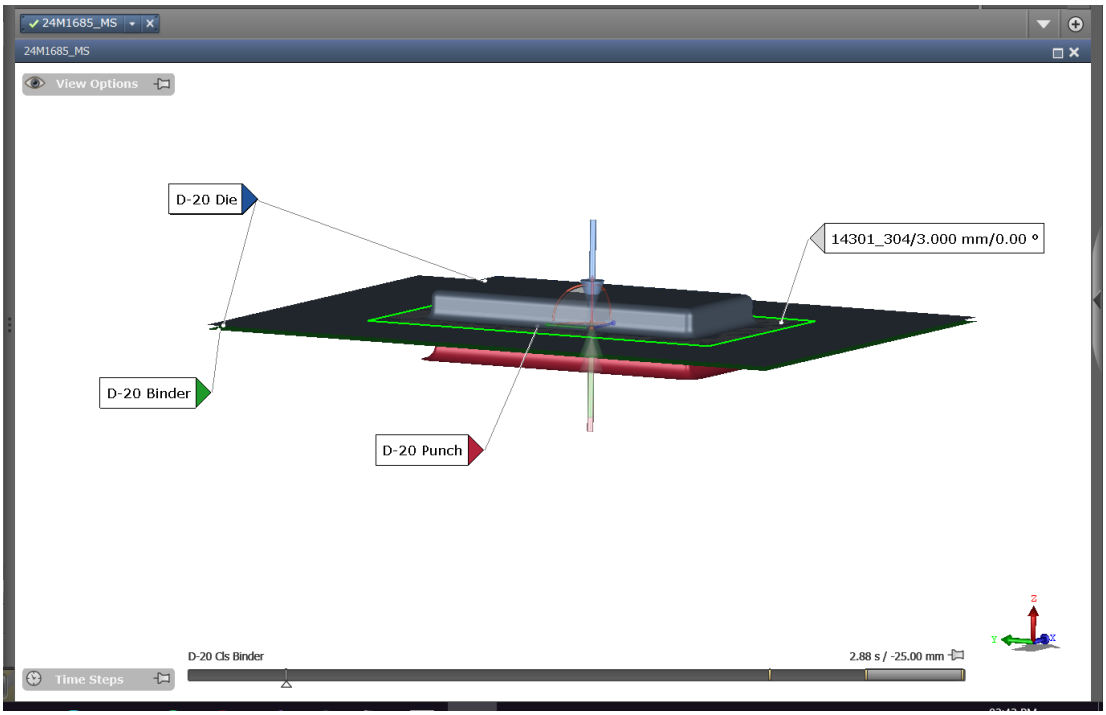


Figure 9 – Complete setup with Punch , Die and Binder

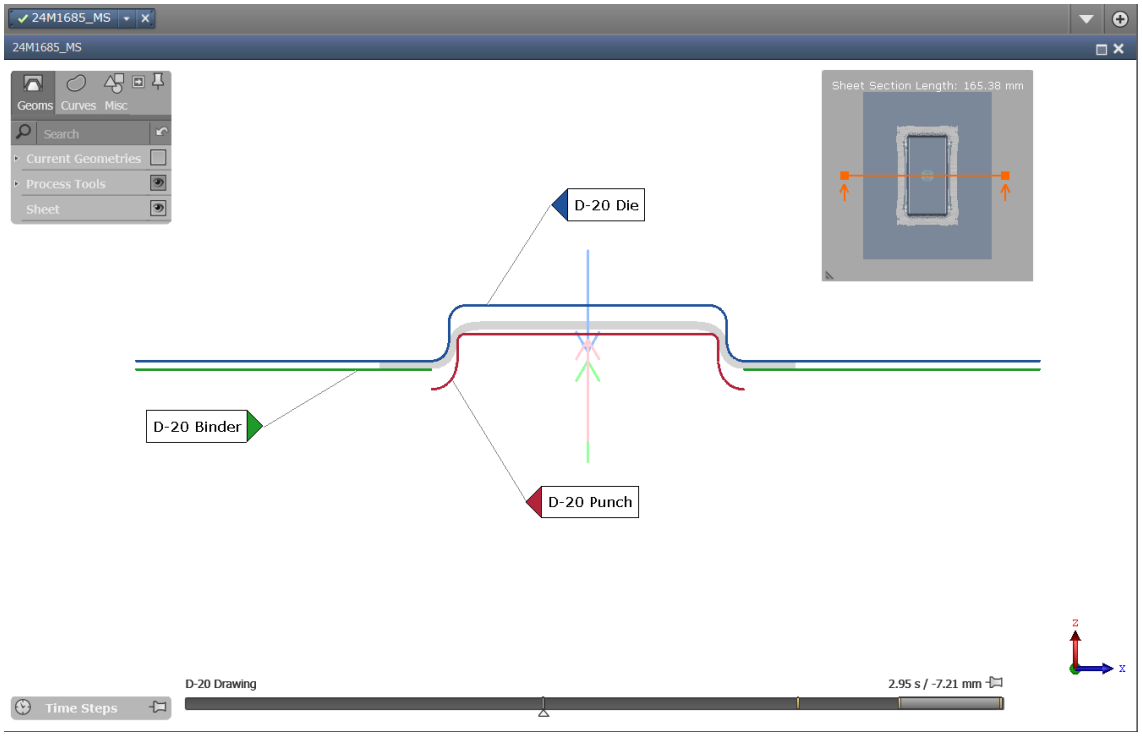


Figure 10 – Section view for complete setup

2.3 Results Simulation

2.3.1 Formability - The Auto Form simulation results confirm the tractor foot step exhibits excellent formability with **no tears, wrinkles, compression, or risk of splitting** during the forming process. The forming limit diagram (FLD) shows all strain values safely within the acceptable range, while thinning analysis demonstrates uniform material flow within permissible limits (-0.2 to 0.2).

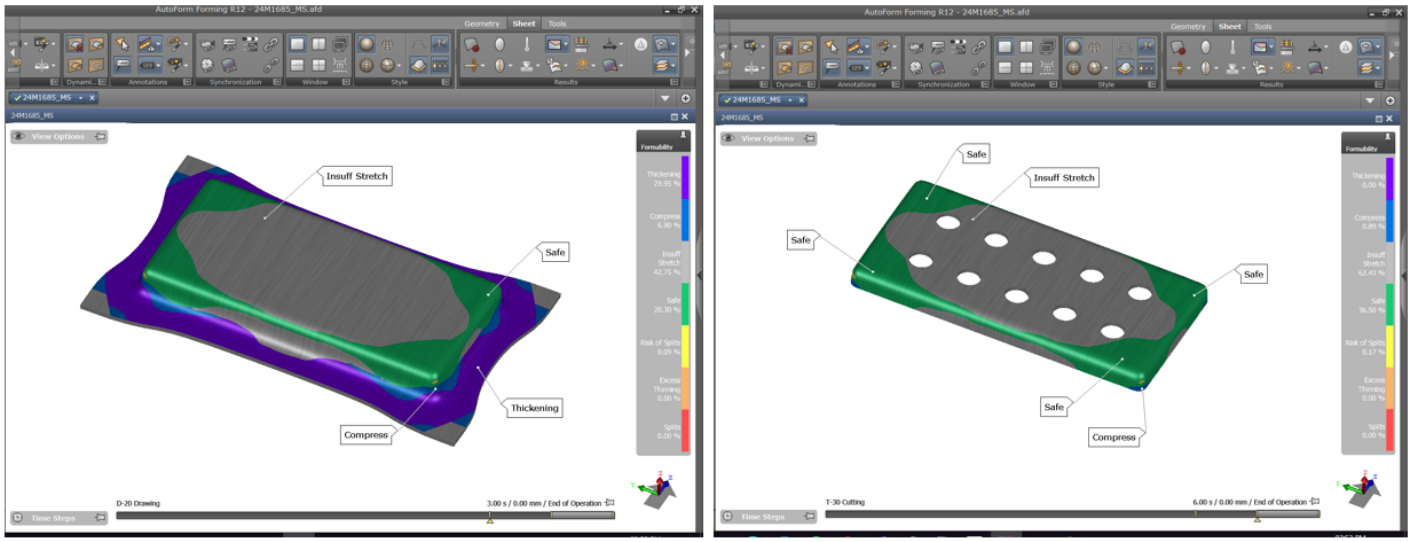


Figure 11 – Formability analysis

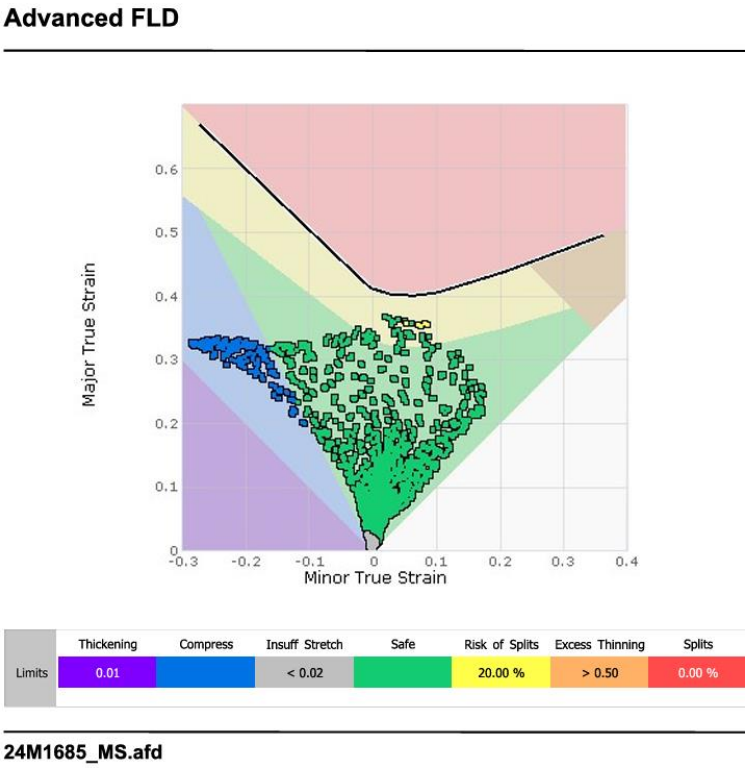
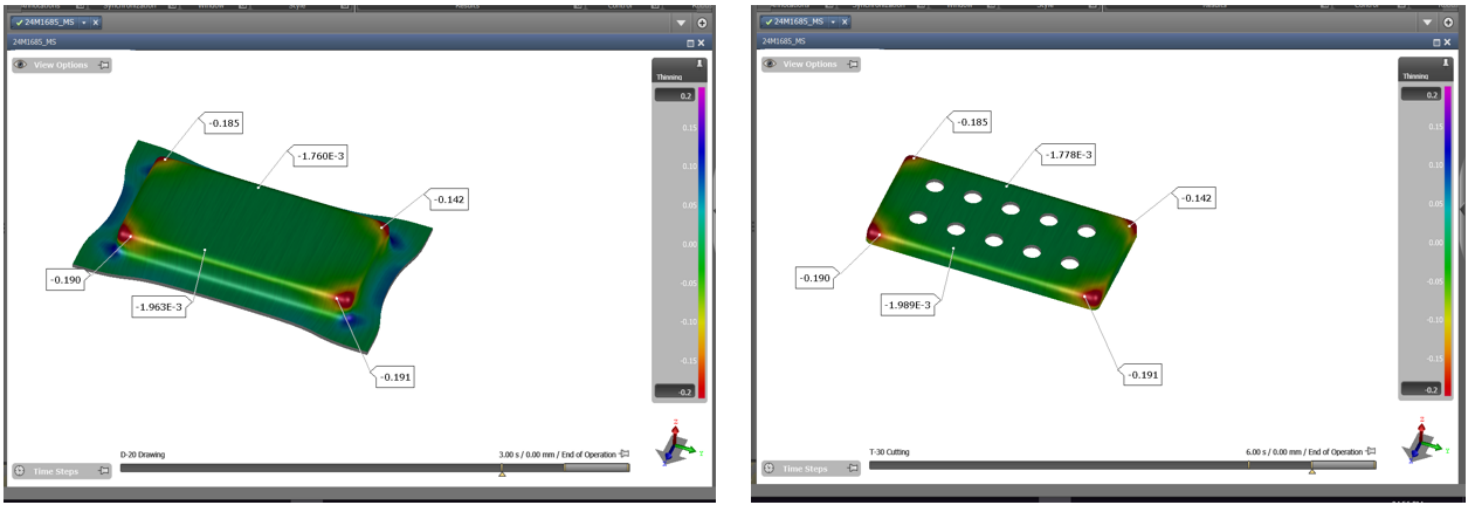


Figure 12 – FLD (in safe zone)

2.3.2 Thinning - The thinning analysis conducted in Auto Form confirmed that material thickness variation remains well within the **acceptable range of -0.2 to 0.2 throughout** the formed foot step component. This result indicates optimal material flow during the stamping process, with no areas exceeding critical thinning limits that could compromise part strength or durability. The uniform thickness distribution validates the selected 3mm S355 steel blank and tooling design as suitable for production, ensuring consistent structural integrity in the final product while maintaining required safety margins against excessive thinning or potential failure points.



End of drawing **End of Cutting**

Figure 13 – Thinning Analysis

2.3.3 Spring back – The spring back analysis revealed dimensional deviations well within the acceptable range of **-0.5mm to 0.5mm** across all critical features of the foot step. This minimal spring back magnitude confirms the effectiveness of the selected material (S355 steel) and forming process parameters, ensuring the component maintains geometric accuracy after tool release. The results indicate that no additional compensation or tooling modifications are required, making the design production-ready while meeting strict dimensional tolerance requirements for tractor applications.

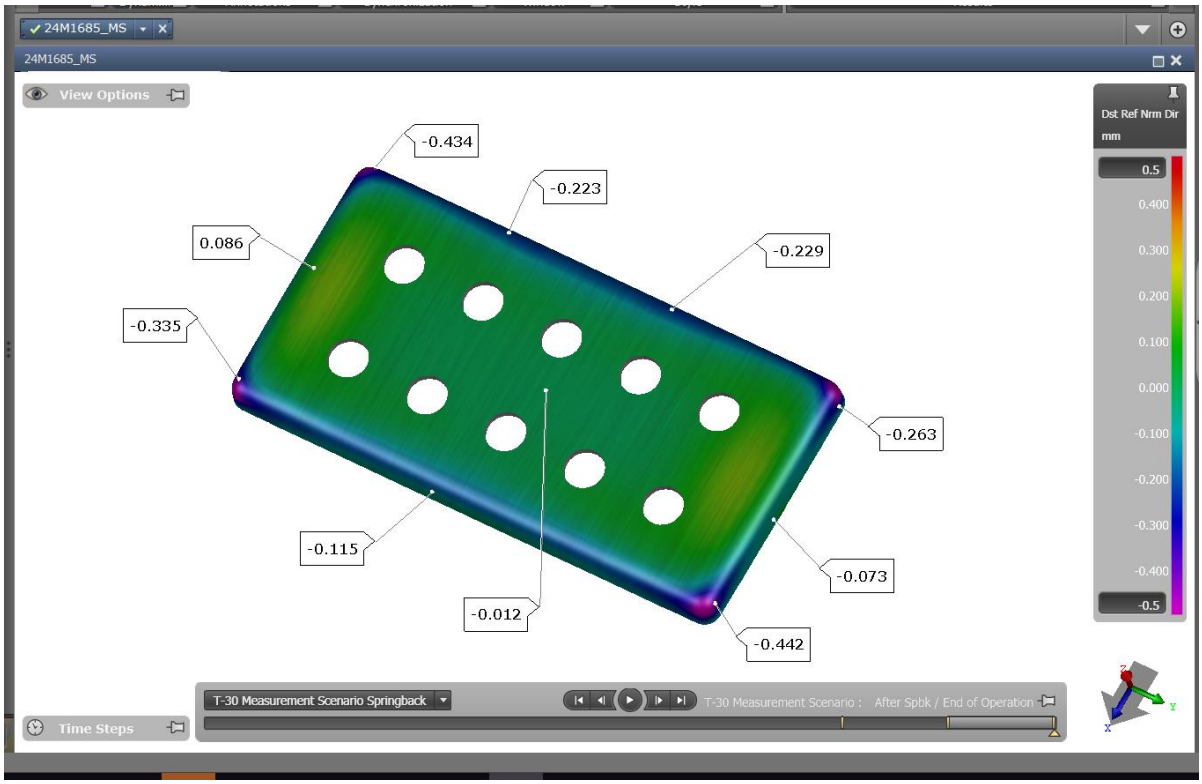


Figure 14 – Spring back Analysis

2.3.4 Wrinkle Analysis – The wrinkle analysis confirmed excellent material stability, with **no wrinkles detected** across the entire foot step component, as evidenced by the complete green zone in the simulation results. This outcome validates that the optimized blank holder force and binder pressure effectively prevented any out-of-plane deformations, ensuring smooth surface quality and dimensional accuracy in the final product. The absence of wrinkles demonstrates robust process control and proper material flow throughout the forming operation.

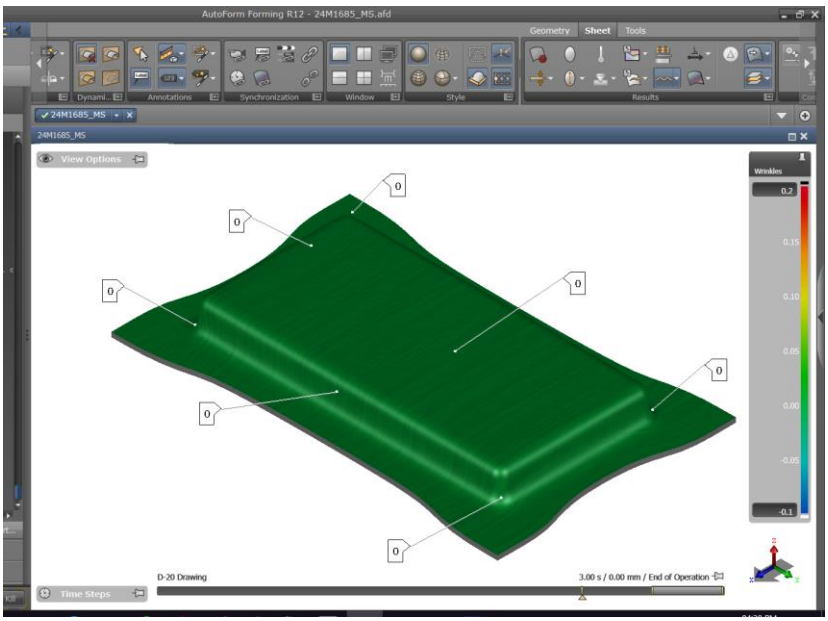


Figure 15 – Wrinkle Analysis

2.3.5 Major Strain Analysis - The major strain distribution analysis of the tractor foot step revealed a global maximum strain of **0.263** and global minimum strain of **1.357E-3**, indicating uniform material deformation during the forming process. The maximum strain value remains well below the forming limit, confirming no risk of localized thinning or fracture. The extremely low minimum strain demonstrates consistent material flow without underutilized regions. These results validate that the component maintains structural integrity throughout the stamping operation, with all strain values falling safely within the acceptable range for S355 steel material. The balanced strain distribution further confirms the effectiveness of the selected blank holder force and lubrication conditions in achieving optimal formability.

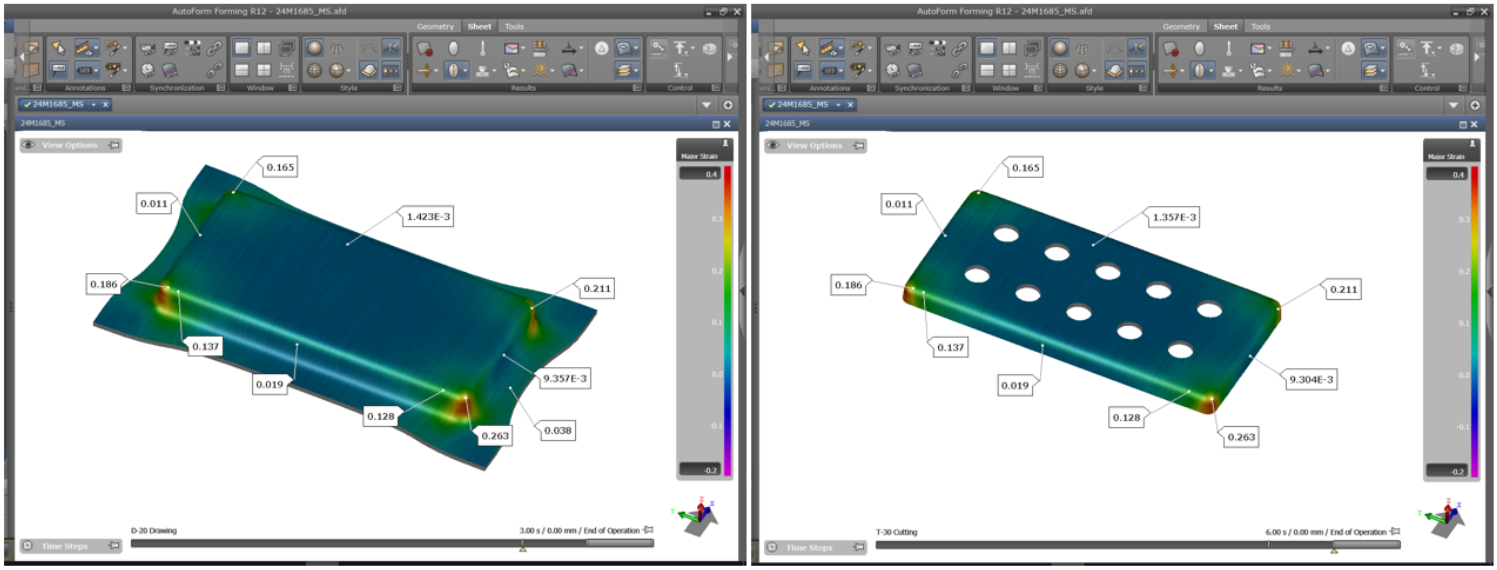
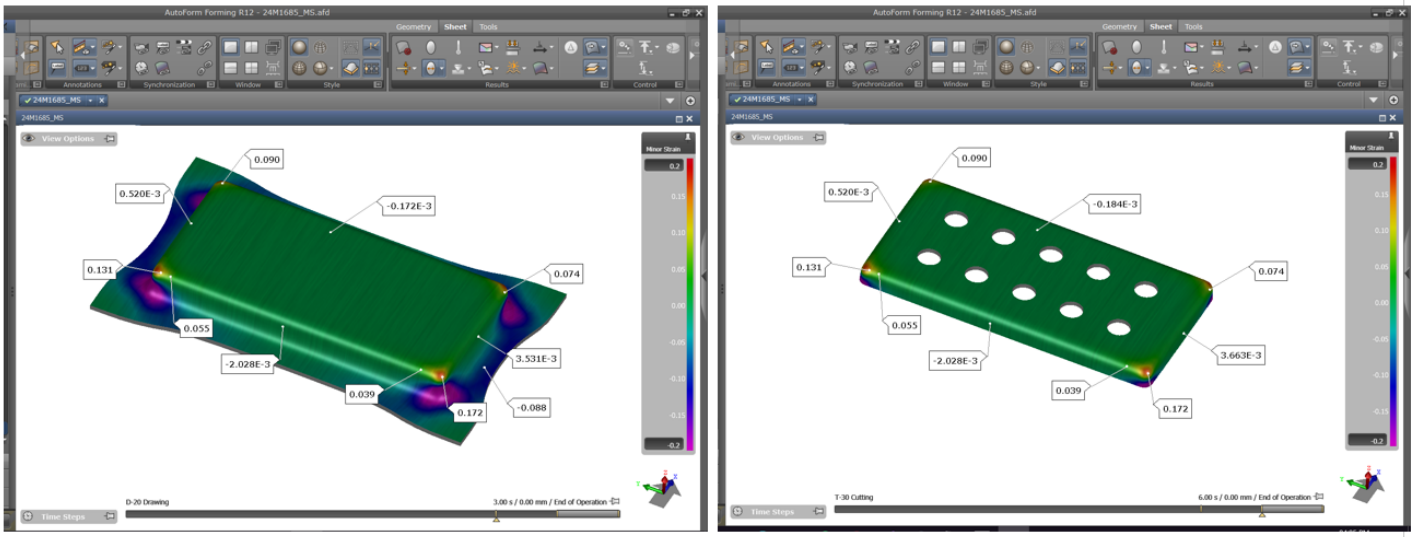


Figure 16 – Major Strain Analysis

2.3.6 Minor Strain Analysis - The minor strain distribution analysis revealed a global maximum strain of **0.172** and global minimum strain of **-2.028E-3**, both well within the safe forming limits. The predominantly positive minor strain values indicate compressive deformation during forming, while the minimal negative value suggests localized tensile effects are negligible. This uniform strain distribution (Figure X) confirms balanced material flow without wrinkles or excessive thinning, validating the robustness of the tool design and blank holder force optimization. The absence of sharp strain gradients further demonstrates stable forming conditions throughout the process.



End of drawing **End of Cutting**

Figure 17 – Minor Strain Analysis

2.4 Final Auto form analysis Result Summary

1. **Formability**: Entire part within FLD safe zone - no risk of splits or cracks
2. **Thinning**: Perfectly controlled (-0.2 to 0.2) - maintains structural integrity
3. **Spring back**: Minimal (±0.5mm) - ensures dimensional accuracy
4. **Wrinkles**: Zero defects - optimal material flow achieved
5. **Major Strain**: Safe range (0.263 max / 1.357E-3 min) - uniform distribution
6. **Minor Strain**: Balanced (0.172 max / -2.028E-3 min) - no critical deformations
7. **Material Behavior**: All parameters confirm production-ready design

Conclusion: The foot step design successfully meets all forming requirements for manufacturing.

3. Design optimization / changes

3.1 Challenges:- The initial footstep design presented critical manufacturability challenges, including **stress concentrations at sharp edges** and **non-uniform thickness distribution** in curved sections, both of which risked forming defects during production. To address these issues, we implemented two targeted improvements: (1) replacing all sharp edges with optimized 2-5mm radii to eliminate stress risers and improve material flow, and (2) redesigning curved surfaces to ensure consistent wall thickness. Post-optimization AutoForm simulations confirmed these changes successfully resolved the thinning variations (±0.2mm tolerance) and stress concentrations while maintaining structural performance, ultimately achieving a production-ready design with reliable formability and consistent quality.

3.2 Mitigation of Challenges:- To address the identified forming issues, key design modifications were implemented. **Sharp edges were eliminated** by incorporating smooth transitional radii at all critical intersections, significantly reducing stress concentrations and improving material flow during stamping operations. Simultaneously, **uniform thickness distribution** was achieved through geometric optimization of curved surfaces, particularly at high-strain regions, ensuring consistent material deformation within the safe thinning limits. These iterative refinements, validated through subsequent Auto Form simulations, resolved earlier formability concerns while maintaining the component's structural performance and ergonomic requirements. The optimized design now demonstrates stable forming behavior with no risk of splits or excessive thinning.

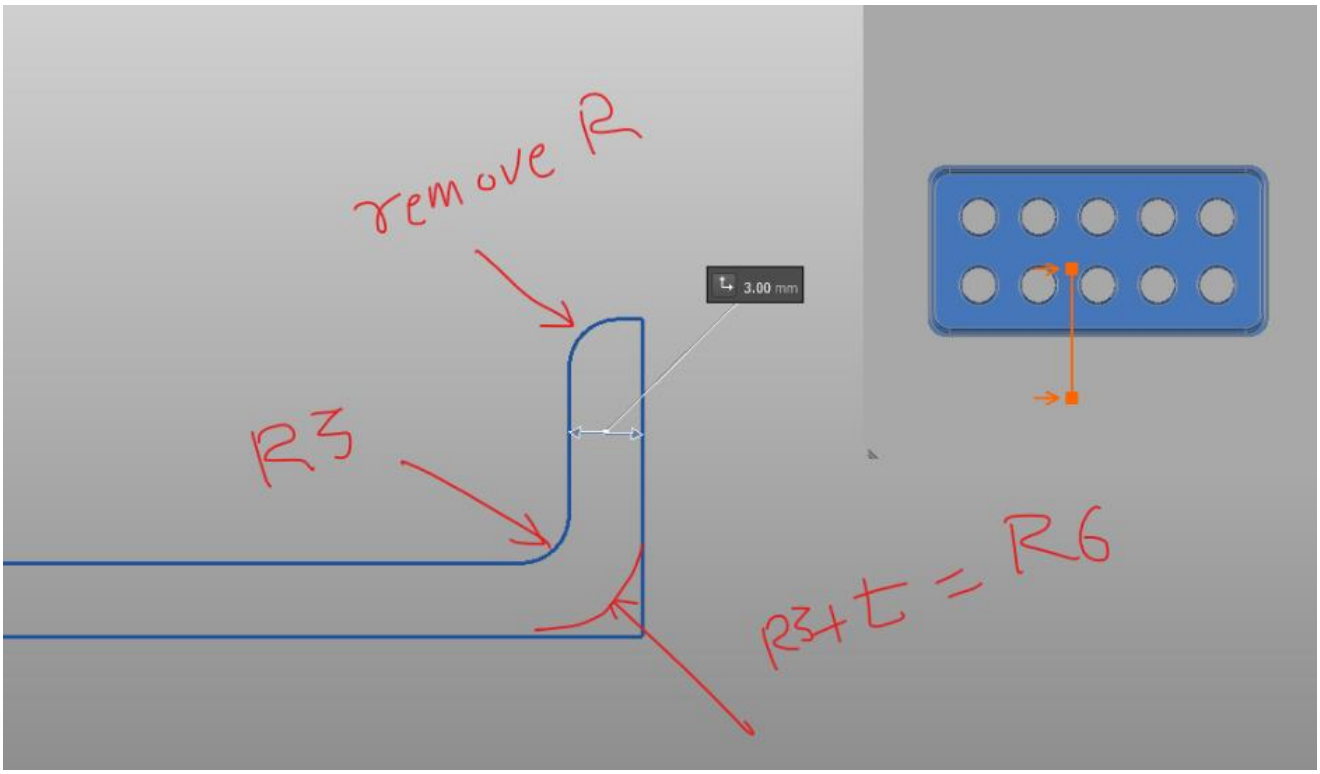


Figure 18 – Challenge of uneven bends

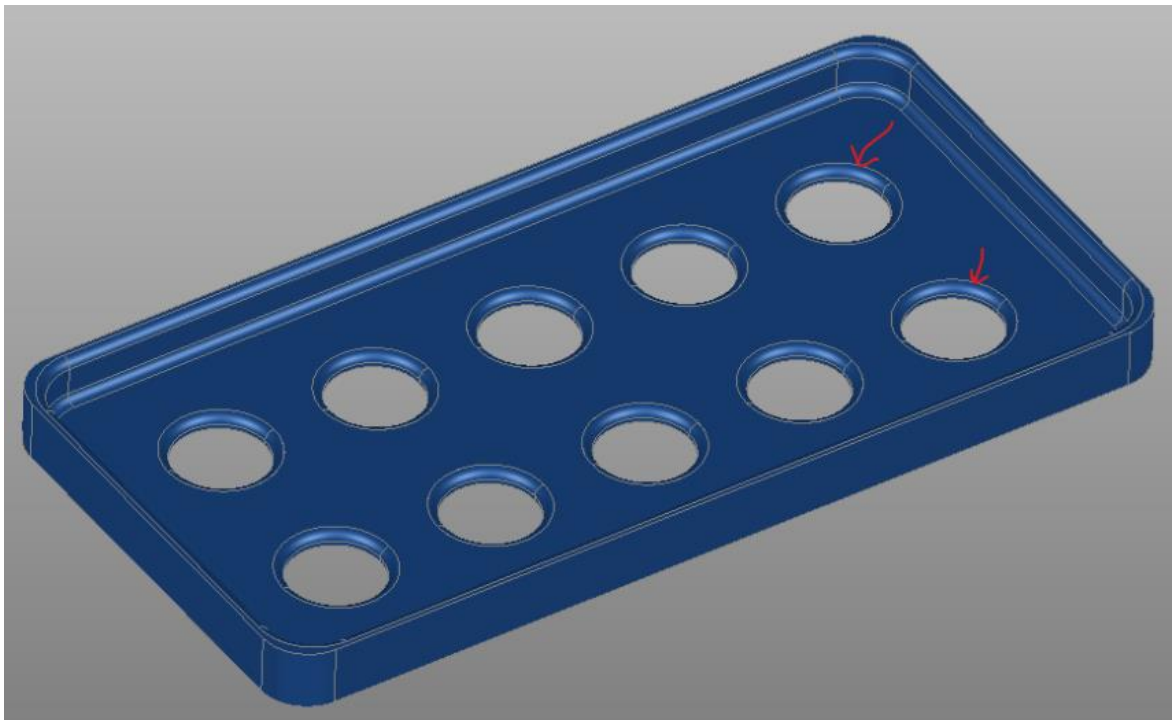


Figure 19 – Removal of round on the hole feature

4. Summary and Learnings

This project successfully demonstrated the design, simulation, and optimization of a tractor footstep using AutoForm software, covering all critical aspects of sheet metal engineering. Beginning with material selection (S355 steel per ISO 1052), the study validated the part's functional requirements—ergonomics, anti-slip design, and durability—through detailed engineering analysis. The AutoForm process setup, including punch, die, and binder configuration, was meticulously simulated to evaluate formability, thinning (±0.2mm), springback (±0.5mm), and wrinkle formation. Results confirmed uniform strain distribution (major strain: 0.263, minor strain: 0.172) with no defects, while iterative design refinements (e.g., edge radius using and thickness optimization) resolved initial stress concentrations. The project underscores the importance of digital validation in achieving first-time-right manufacturing for sheet metal components.

Through this hands-on project, we gained practical expertise in sheet metal forming simulation, learning how to configure Auto Form's tooling setup (punch, die, binder), interpret critical results like FLD, thinning, and spring back, and validate material behavior (S355 steel) against industrial standards. The software enabled us to identify and resolve real-world forming issues—such as stress concentrations from sharp edges and uneven thickness distribution—through iterative design improvements. Most importantly, we developed the ability to translate simulation data into manufacturable solutions, understanding how process parameters (blank holder force, lubrication) impact formability. This experience bridged theoretical knowledge with industrial practice, emphasizing simulation's role in reducing prototyping costs and ensuring defect-free production.